

Phase 9: Adviser Redaction Review Gate & Failsafe Egress Control

1. Executive Summary & Privacy Invariant

The Adviser Redaction Review Gate is the core technical and operational control upon which the entire privacy architecture of Case Ace v2.0 rests (implementing Constraints C1, C4, C5, and C9).

Under no circumstance is raw audio, client PII, or unstructured consultation text transmitted to external cloud processors (Google Cloud Speech-to-Text v2 or Vertex AI Gemini 1.5 Pro). Before any data packet departs the local device:

All local acoustic Pass 1 ASR detections and Layer 1–3 identifier classifications are presented to the adviser in volatile RAM.

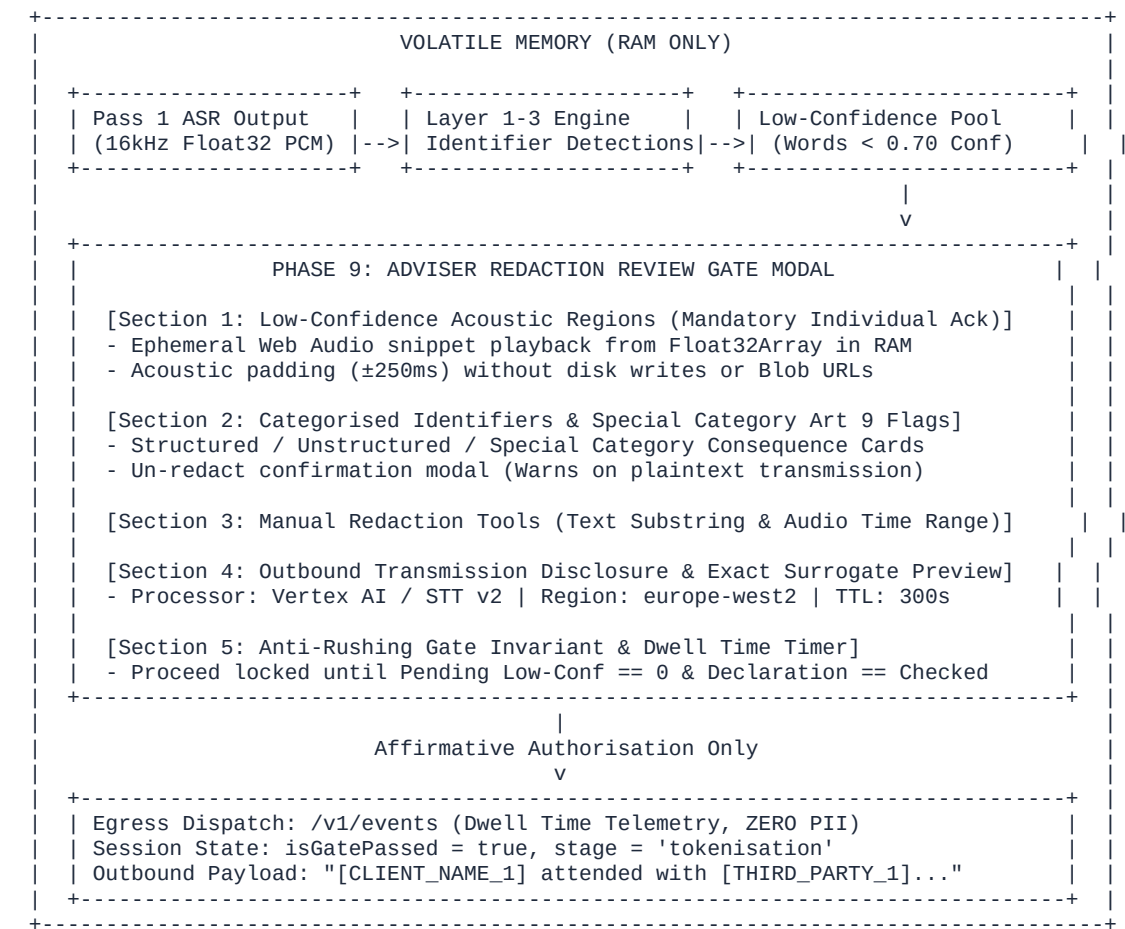
Every acoustic region where Pass 1 ASR confidence fell below 0.70 (70%) is isolated in a prominent escalation section and must be individually auditioned and acknowledged.

The adviser has full agency to add manual redactions (by text or audio time range) or un-redact false positives with a mandatory plaintext disclosure warning.

The system presents a transparent outbound transmission statement detailing the cloud processor, designated region (europa-west2 London, UK), ephemeral credential lifetime (300s), and the exact surrogate-tokenised payload preview.

The proceed control is strictly locked until all low-confidence regions are acknowledged and the affirmative declaration is checked.

2. Technical Architecture & Components



3. Core Modules

3.1 Audio Snippet Player (client/src/audio/audioSnippetPlayer.ts)

Direct Volatile RAM Playback: Directly instantiates AudioBuffer and AudioBufferSourceNode via Web Audio API.

Zero Disk Writes: Never creates Blob, ObjectURL, or temporary cache entries.

Contextual Acoustic Padding: Applies 250ms–300ms pre/post padding so advisers hear full acoustic context for mumbled or co-articulated words.

Immediate Resource Cleanup: Stops and releases previous sources on item transition.

3.2 Redaction Gate Manager (client/src/redaction/redactionGateManager.ts)

Acoustic Low-Confidence Pool: Aggregates tokens with ≤ 0.70 confidence from localAsrResult.lowConfidenceWords and detectedIdentifiers.

Readiness Verification (checkGateReadiness): Returns canProceed: false and explicit blocking reasons whenever pendingCount > 0.

Outbound Transmission Disclosure (getOutboundDisclosure): Dynamically computes the exact surrogate-tokenised text string about to depart the browser, along with token counts and region disclosures.

Failsafe Proceed Handler (executeAffirmativeProceed): Fails closed if affirmative consent is absent or pending items exist; records dwell time and emits security telemetry.

3.3 Interactive Modal UI (client/src/components/RedactionReviewGateModal.tsx)

High-Risk Low-Confidence Section: Visual amber banner with individual card for each low-confidence token, audio playback trigger, and individual acknowledge toggle.

Categorised Identifier Review: Displays detected entities grouped by layer with surrogate tags, audio players, and un-redact confirmation warnings.

Manual Redaction Controls: Enables advisers to redact missed text or specify start/end seconds for acoustic audio bleeps.

Live Outbound Disclosure Box: Provides side-by-side verification of what is redacted vs retained.

Anti-Rushing Proceed Bar: Disables proceed button until all conditions are met; displays active gate dwell timer.

4. Anti-Rushing Design & Dwell Time Telemetry

Advisers operating between consultations face time pressure. Case Ace specifically defends against habituation and accidental bypass:

No “Redact All” Shortcut: A single global bypass button is intentionally absent from the codebase.

No Cross-Session Preference Caching: Review decisions and low-confidence acknowledgements are strictly ephemeral; nothing is written to localStorage or indexed across sessions.

Mandatory Affirmative Checkbox: Requires active click on declaration statement.

Active Gate Dwell Time Measurement:

$\Delta t = t_{\text{completed}} - t_{\text{opened}}$

The dwell time is logged to the backend /v1/events endpoint in JSON format:

```
{
  "type": "redaction_gate_completed",
  "timestamp": "2026-09-02T08:04:44.000Z",
  "details": {
    "dwellTimeMs": 14250,
    "lowConfidenceReviewedCount": 3,
    "manualAddedCount": 1,
    "manualRemovedCount": 0,
    "totalTokenCount": 5
  }
}
```

Note: Telemetry payloads are strictly sanitized and never contain consultation text, transcripts, audio, or client PII.

5. Verification & Acceptance Criteria

All 5 Phase 9 requirements were formally tested and verified via scripts/test-phase9.mjs:

- ' Criterion 1: Zero data egress before affirmative gate authorisation.
- ' Criterion 2: Individual acknowledgement required for all low-confidence acoustic regions (≤ 0.70).

- ' Criterion 3: Adviser ability to add manual redactions and remove false positives with plaintext warning.
- ' Criterion 4: Strict absence of bypass/skip paths across all code branches.
- ' Criterion 5: Accurate gate dwell time captured and dispatched via security telemetry.